

SECTION E: MODEL EVALUATION, INTERPRETABILITY, AND ACTUARIAL FAIRNESS REVIEW

MEMORANDUM

- **TO:** MSAC512 Assessment Board & Lead Actuarial Examiner
- **FROM:** Trainee Data Analyst
- **DATE:** 23 September 2026
- **SUBJECT:** Residual Analysis, Feature Importance Decomposition, Local SHAP Explanations, and Actuarial Fairness Evaluation

E.1 Residual Analysis & Systematic Mis-Pricing Patterns

Residual diagnostics were conducted on the holdout test set ($n = 37$) for the selected final **Random Forest Regressor** model. The test set yielded a positive **Mean Residual of** +\$105,350.75, indicating a systematic tendency of the model to under-predict property valuations across the market.

$$\text{Residual}_i = y_i - \hat{y}_i = \text{Price}_i - \hat{\text{Price}}_i$$

- **Residuals vs. Fitted Values (Heteroscedasticity):** Residuals stay reasonably centered for lower-to-mid value properties ($\leq \$400,000$). However, for high-end properties ($> \$800,000$), residuals fan out significantly into positive territory. Because tree ensembles average leaf node targets, extreme luxury valuations are pulled back toward the dataset mean.
- **Residuals vs. Stand Size (m^2):** Since land area dominates prediction weight, large stand sizes ($> 2,000 \text{ m}^2$) exhibit the largest positive residual variances, confirming non-linear scaling premiums for expansive plots that linear combinations fail to fully capture.
- **Residuals vs. Suburb:** Prime suburban locations (e.g., Borrowdale, Mount Pleasant) show positive mean residuals (+\$150,000 to +\$300,000), whereas high-density suburbs (e.g., Chitungwiza, Ruwa) exhibit near-zero or small negative residual means, confirming that under-prediction is concentrated in upmarket tiers.

E.2 Global Feature Importance & Local SHAP Explanations

A. Global Permutation Feature Importance Ranking

Evaluating permutation importance drops in holdout R^2 score reveals extreme reliance on physical dimensions:

Rank	Feature Name	Category	Permutation Importance (ΔR^2)	Relative Weight (%)
1	log_stand_size	Size & Capacity	0.2108	95.2%
2	bathrooms	Size & Capacity	0.0035	1.6%
3	bedrooms	Size & Capacity	0.0018	0.8%
4	strat_suburb_Borrowdale	Location / Spatial	0.0031	1.4%
5	strat_suburb_Chitungwiza	Location / Spatial	0.0022	1.0%
6	img_vis_condition_score	Condition & Finish	0.0000	0.0%
7	txt_finish_luxury_flag	Condition & Finish	0.0000	0.0%

B. Local SHAP Explanations for Individual Predictions

Local Explanation 1: Property ID #104 (Borrowdale Suburban Compound)

- Actual Price: \$850,000
- Model Fitted Value: \$582,240
- Base Mean Value: \$320,000

Base Output Value = \$320,000
 +\$235,000 (Size: log_stand_size = 8.29 / 4,000 m²)
 +\$18,400 (Size: bathrooms = 4)
 +\$8,840 (Location: strat_suburb_Borrowdale = 1)
 +\$0 (Condition / Visual Inputs: Negligible effect)
 Final Prediction = **\$582,240** (Residual: + \$267,760 Under-prediction)

Local Explanation 2: Property ID #218 (Chitungwiza Starter Property)

- **Actual Price:** \$45,000
- **Model Fitted Value:** \$48,200
- **Base Mean Value:** \$320,000

Base Output Value = \$320,000
 −\$258,000 (Size: log_stand_size = 5.30 / 200 m²)
 −\$8,800 (Location: strat_suburb_Chitungwiza = 1)
 −\$5,000 (Size: bedrooms = 2)
 Final Prediction = **\$48,200** (Residual: − \$3,200 Over-prediction)

E.3 Predictive Power Attribution & Feature Group Decomposition

Aggregating feature importance directly yields the required decomposition input for Section F:

Total Predictive Power = 100%

Feature Category	Key Included Variables	Attribution (%)
1. Size & Capacity	log_stand_size, bedrooms, bathrooms	97.6%
2. Location / Spatial	Suburb One-Hot Indicators (Borrowdale, etc.)	2.4%
3. Condition & Finish	Visual condition scores, text luxury flags	0.0%
TOTAL	All Model Features	100.0%

- **Core Actuarial Finding:** Physical land area and room counts drive virtually all predictive capacity (97.6%). Unstructured text and image inputs provided 0.0% net marginal gain in this sample, likely due to feature sparsity or noise in automated visual embeddings.

E.4 Actuarial Fairness, Proxy Discrimination, & Regulatory Review

A. Suburb as a Proxy for Sensitive Characteristics

In Zimbabwean property markets, geographical location serves as a strong correlate for socioeconomic status and demographic background:

1. **Proxy Discrimination Risk:** High-density townships (e.g., Chitungwiza, Highfield) heavily correlate with lower average household incomes and informal employment sector participation. If an Automated Valuation Model (AVM) places heavy negative loadings on location tokens, it risks entrenching redlining—restricting mortgage credit access and suppressing property asset growth in developing areas.
2. **Low Empirical Spatial Bias:** However, our feature attribution demonstrates that location accounts for only 2.4% of model predictive power, while objective property size accounts for 97.6%. This minimizes the risk of location acting as an unfair discriminatory proxy.

B. Regulatory Defense & Justification Strategy

If questioned by regulators (such as the Reserve Bank of Zimbabwe or IPEC) regarding location loadings:

1. **Empirical Evidence of Minimal Reliance:** The actuarial decomposition proves that the model does not rely on location as a primary valuation driver (2.4% total impact). Valuation differentials are driven by physical land size (97.6%) rather than spatial stereotyping.
2. **Infrastructure Risk Adjustment:** Location variables capture objective local infrastructure variations—such as municipal water provision, road access, and electrical grid reliability—which directly affect structural maintenance costs and market resale liquidity.